

Armin Lotfy

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Professional Summary

Ph.D. Electrical Engineer and AI/Control specialist building **Python-based, data-driven and adaptive control** for **partially observable** and safety-critical systems. Expert in **RL/DL (PPO, SAC, TD3)**, **EKF/UKF**, **MPC**, and **stochastic optimization** (GA/PSO), with production-grade implementations in **TensorFlow**, **RLLib**, and **MATLAB/Simulink**. Design **POMDP** formulations, learnable controllers, and **digital-twin** simulations; apply **time-series forecasting, and multivariate statistics** to close the loop from **ingest, model, optimize, deploy**. Comfortable with **GPU-aware training**, **Optuna** library, reproducible pipelines, and **Git/GitHub CI**. Familiar with **agentic AI** (LLMs, tool use, multi-agent orchestration) and workflow automation (e.g., **n8n**) to accelerate research and ops.

Professional Experience

09/2020 – 09/2025
Ottawa, Ontario

Research Assistant

Carleton University

- **Formulating** multi-objective, partially observable control problems as **POMDPs** with hard/soft constraints; **defining objective vectors** (performance, safety, cost, degradation) and **using weighted-sum and ϵ -constraint** methods.
- **Modeling** electric vehicles, batteries, and **hybrid energy storage systems** in **Python**.
- **Implementing** plant-controller stacks in **MATLAB/Simulink** (Simscape when applicable) and **Python**; **building** digital-twin testbeds and **scenario generators**.
- **Designing and deploying PPO/SAC/TD3** actor-critic policies (feed-forward & LSTM) with **reward shaping, safety filters/shields, domain randomization**, and **offline-to-online RL**; **training** with **Ray/RLLib** and **TensorFlow/Keras**, and **tuning** with **Optuna**.
- **Building** predictive models for **dynamic state estimation with data-driven inference** to improve accuracy and responsiveness.
- **Applying MPC**, stochastic/robust optimization, and **metaheuristics (GA/PSO)**, and **designing** controller fallbacks and **safe-degradation modes** for real-time deployment.
- **Implementing, simulating, and validating** solutions in **Python (TensorFlow/Keras)** and **MATLAB/Simulink**.

09/2022 – 04/2024
Ottawa, Ontario

Teaching Assistant

Carleton University

- Mentored 600+ students across Mechatronics & Circuits (ECOR 1043/1044, Python) and Switching Circuits & Digital Design (ELEC 2607, VHDL).
- Led labs and project coaching, guiding implementations on Raspberry Pi and Xilinx FPGA platforms.

09/2022 – Present
Ottawa, Ontario

Technical Support Clerk and Exam Site Facilitator

Carleton University

- **Providing** frontline technical support for students/faculty—**diagnosing and resolving** HW/SW issues, **mentoring** exam-tool usage, **documenting** fixes; **triaging** requests and **escalating** complex cases for timely, clear resolution.
- **Coordinating** multi-site exam logistics (proctors/instructors, room setup, compliance/accommodations); **managing** real-time incidents in high-volume settings, **minimizing** disruption and **keeping** sessions on schedule.

Skills

Technical Skills

- **Machine Learning algorithms:**
K-means Clustering, Regression Trees, Principal Component Analysis (PCA), Predictive Modeling, Time Series Forecasting
- **Reinforcement Learning & Deep Learning algorithms:**
TD3, MATD3, DDPG, PPO, MAPPO, SAC, CNN, RNN, LSTM
- **Programming & Frameworks:**
MATLAB, Simulink, Python (NumPy, SciPy, Pandas, Matplotlib), TensorFlow, RLlib, Stable-Baselines3, Git, Docker, LaTeX, VS Code, Linux Bash, VHDL
- **Large Language Models:** LLMs, RAG, Fine-tuning, n8n, Ollama

Projects

RL-Based Energy Management System for BEVs — IRES Lab

- **Optimize power distribution and efficiency under variable loads** by developing a MATLAB/Simulink Energy Management System powered by multi-agent reinforcement learning.
- **Drive SOC stability, efficiency, and battery longevity** by formulating a continuous-state/action MDP with reward shaping around SOC deviation, efficiency, and degradation.
- **Ensure stable, deployable control** by training two TD3 agents (power-flow control, SOC balancing) with target-policy smoothing and delayed updates, validated in closed-loop simulations for real-time-viable performance.

CTDE-Based Multi-Agent BMS for SOC/SOH Equalization — IRES Lab

- **Optimize current allocation while preserving battery health** by applying PPO in MATLAB/Simulink to a multi-objective control problem, using weighted scalarization and constraint penalties; validated via closed-loop simulations.
- **Coordinate safe, efficient behavior under uncertainty** by designing a shared-reward POMDP with belief-state tracking to handle sensor noise and enable joint optimization of efficiency, safety, and degradation.

POMDP Controller for BEVs with Battery–Supercapacitor HESS (Ray/RLlib) — IRES Lab

- **Enable uncertainty-aware policy learning** by formulating control as a **POMDP** with belief-state updates from noisy sensors; train PPO using history stacks and predictive features to handle partial observability.
- **Deliver real-time SOC/SOH forecasting and estimation** via **CNN-LSTM** predictors and an EKF for online inference; implemented in Python (**RLlib, TensorFlow 2, Gymnasium**) and validated in closed-loop with RMSE/MAE for deployment readiness.

Enhancing Power Sharing in Microgrids - IRES Lab

- **Optimize power sharing and grid stability** by designing a Proximal Policy Optimization (PPO)–based controller for microgrids that allocates power among distributed resources under dynamic load and generation.
- **Enable decision-optimal control in simulation** by modeling the problem as a Markov Decision Process (MDP) in MATLAB/Simulink, defining state/action spaces, constraints, and reward shaping for efficiency, voltage/frequency regulation, and asset longevity.

Education

09/2020 – 09/2025	Ph.D. in Electrical Engineering <i>Carleton University</i>
09/2016 – 12/2019	M.Sc. in Electronic Digital Engineering <i>IUST</i>
09/2010 – 09/2016	B.Sc. in Electronic Engineering <i>BNUT</i>

Publications

- 05/2025 **Centralized Multi-Agent SOC Control for Battery Health Using Proximal Policy Optimization in EVs** [↗](#)
IEEE Transactions on Vehicular Technology
Lotfy, A., Alzayed, M., Chaoui, H., & Boulon, L.
- 07/2024 **Enhancing Energy Management Strategy for Battery Electric Vehicles: Incorporating Cell Balancing and Multi-Agent Twin Delayed Deep Deterministic Policy Gradient Architecture** [↗](#)
IEEE Transactions on Vehicular Technology
Lotfy, A., Chaoui, H., Kandidayeni, M., & Boulon, L.
- 06/2025 **Cutting-Edge EMS Technologies for EVs and HEVs: Recent Developments and Future Directions** [↗](#)
IEEE Access
- 06/2025 **Hierarchic Multi-Agent Energy Management for Extended Driving Range through Battery Cell Balancing** [↗](#)
34th IEEE International Symposium on Industrial Electronics (ISIE 2025)
Lotfy, A., Chaoui, H., Kandidayeni, M., Boulon, L., Khaneghah, M. Z., & Hasheminasab, S.

Awards

- 09/2025 **Senate Medal nominee for Outstanding Graduate Work (Doctoral Level)**
Carleton University
Senate Medal Nominee for Outstanding Graduate Work (Doctoral Level), recognizing exceptional doctoral research and academic achievement; nomination endorsed by the department and submitted to the university's Senate for final consideration.
- 03/2025 **CMC Electronics Inc. Scholarship**
Carleton University
CMC Electronics Scholarship recognizing academic excellence in Electrical & Computer Engineering; conferred by the Faculty of Engineering Scholarships Committee based on departmental recommendation.

Non-Profit Organizations

- 05/2025 – 07/2025 **Member of the Technical Program Committee, IEEE 102nd Vehicular Technology Conference (VTC2025)**
Served as a **Technical Program Committee (TPC) Member** for the IEEE 102nd VTC2025
- 05/2025 – 07/2025 **IEEE Vehicle Power and Propulsion Conference (VPPC) 2025 Reviewer**
Contributed as a Technical Reviewer for IEEE VPPC 2025, an internationally recognized conference in electric mobility and powertrain systems
- 01/2022 – 02/2022 **IEEE Vehicle Power and Propulsion Conference (VPPC) 2022 Reviewer**
Served as a Technical Reviewer for the IEEE Vehicle Power and Propulsion Conference (VPPC) 2022

References

Professional references and referee contact information available upon request.